MIL-C-83957A (USAF)
23 July 1975
SUPERSEDING
MIL-C-83957
25 November 1970

MILITARY SPECIFICATION

CLEANING AND POLISHING PAD, NONMETALLIC (FOR AIRCRAFT)

1. SCOPE

*1.1 Scope. This specification covers two types of nonwoven, nonmetallic cleaning and polishing pads which are nonabrasive. These pads are made of a polyester fiber and are bonded together (uniformly) with an adhesive.

*1.2 Pads shall consist of two types. The type I pad shall have a thickness of 3/8 inch ± 1/16 inch. The type II pad shall have a thickness of 1 inch ± 1/8 inch.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-3816	Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage
MIL-R-25134	Remover, Paint, and Lacquer, Solvent Type
MIL-C-25769	Cleaning Compound, Aircraft Surface, Alkaline Water Base
MIL-C-38334 (USAF)	Corrosion Removing Compound, Prepaint, for Aircraft Aluminum Surfaces

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

*2.2 Other Publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 7920

Source: https://assist.dla.mil -- Downloaded: 202640-23T09:54Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D1117-69 - Testing Nonwoven Fabrics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Materials. Materials used in the manufacture of the pad shall meet the applicable test requirements of this specification.

3.2 Fiber. The fiber of the pad shall be a crimped, cut filament polyester fiber meeting the requirements of Table I and Table II.

3.3 Adhesive. The adhesive shall meet the requirements of Table II.

*3.4 Dimensions and Dimensional Tolerances: The pads shall be made in sheets, 6 x 12 inches, 12 x 12 inches, or 12 x 24 inches. Tolerance on the above dimension shall be ± 1/8 inch. The thickness of the pads shall be as follows:

Type I	Type II
3/8 inch ± 1/16 inch	1 inch ± 1/8 inch

Table I

Requirements for Crimped, Cut Polyester

Property	Value		Test Paragraph
	Type I	Type II	
Material	Polyester	Polyester	None
Denier	6 average	15 average	4.5.4
Crimps per inch *	12 - 14 average, each peak and valley is counted	8 - 12 average, each peak and valley is counted	4.5.4
Staple length	1-1/2 inch	1-1/2 inch	4.5.4
Tenacity, Grams/Denier	3.4 - 4.5 Range	4.0 - 5.0 Range	4.5.4
% Elongation	30 - 70%	30 - 70%	4.5.4

*3.5 Total Weight. The total weight of the type I pad shall be 7.73 ± 0.65 grams per 24 ± 1/4 square inches when tested, as specified in 4.5.3. The total weight of the type II pad shall be 15.9 ± 2.4 grams per 24 ± 1/4 square inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

*3.6 Breaking Strength. The dry breaking strength of both the type I and type II pads shall be 50 ± six pounds per inch of width in the length direction and 38 ± six pounds per inch of width in the width direction when tested, as specified in 4.5.2.

*3.7 Water and Chemical Resistance. When tested, as specified in 4.5.2, the pad shall not show more loss in tensile strength than specified in Table II.

Table II

Water and Chemical Resistance

TEST SOLUTION	Maximum Allowable LOSS IN TENSILE STRENGTH - %		TEST PARAGRAPH
	TYPE I	TYPE II	
Water	7	10	4.5.2
5% NaOH	76	50	4.5.2
Mineral Spirits 165° BP	10	20	4.5.2
* Mixture of 1 Part MIL-C-25769 Cleaning Compound Alkaline Waterbase and 3 parts water	25	20	4.5.2
Corrosion Removing Compound (MIL-C-38334)	25	35	4.5.2
Paint Remover MIL-R-25134	50	90	4.5.2

3.8 *Workmanship.* All details of workmanship shall be in accordance with high-grade manufacturing practices for the type of product specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of Test. All the tests required herein for the testing of the pads are classified as acceptance tests, for which necessary sampling techniques and methods of testing are specified in this section.

4.3 Sampling for Lot Acceptance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

Table III

Immersion Tests

TEST SOLUTION	TEMPERATURE OF TEST SOLUTION	TIME (MIN)
Water	Boiling (212° F)	10
5% by Weight NaOH	150 ± 5° F	5
165° Boiling Point Mineral Spirits	73 ± 5° F	5
Paint Remover per MIL-R-25134	73 ± 5° F	10
Mixture of One Part Water Base Cleaner per MIL-C-25769 and Three Parts Water (by Volume)	73 ± 5° F	15
Corrosion Removing Compound, MIL-C-38334	73 ± 5° F	15

Table IV

Classification of Visual and Dimensional Defects

CATEGORIES	DEFECTS
Critical	None defined
Major:	
101	Wrinkles in material
103	Excessively large hard particles of adhesives within critical working area
104	Torn edges
105	Gouged surfaces or excessively large thin areas
106	Excessively heavy or light weight areas
107	Flat, matted (low loft) areas
108	Irregular shape or sheets (not rectangular or as specified)
* 109	Mats not within dimensional tolerance
Minor	None specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

4.3.1 Lot. For the purpose of inspection, a lot shall consist of not more than 100 pounds of material of the same type, class, and dimensions manufactured from the same batch for delivery at one time by one supplier.

*4.3.2 Samples. A sample from each lot shall be selected in accordance with MIL-STD-105, using an AQL of 4.0 percent defective. A sample shall consist of 144 square inches of pad. Each sample shall be suitably tagged to identify it with the lot represented. Samples shall be subjected to tests specified herein.

*4.4 Examination for Defects. Each of the sample pads selected in accordance with 4.3.2 shall be examined visually and dimensionally to verify compliance with this specification. Examination shall be conducted as specified in Table IV. Any sample containing one or more defects shall be rejected and if the number of defective indicators in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Tests.

4.5.1 Thickness Measurement. The thickness measurement shall be taken by placing a 4 x 6 inch sample between the centers of two approximately 4 x 6 x .045 inch aluminum plates and measuring the thickness with a micrometer. No pressure is to be exerted on the micrometer. The average of at least three measurements shall be reported.

*4.5.2 Water and Chemical Resistance. Samples for this test may be taken from the sample quantity selected in accordance with 4.3.2. Three samples shall be designated as control samples and thus as test samples for each of the solutions specified in Table III. Each set of three test samples shall be completely immersed in the applicable test solution at the temperature and time specified in Table III. Tensile strength shall be run on the test samples and the control samples using procedure specified in ASTM D1117-69, with exception that the test specimen is 2 x 7 inches. The average tensile strength for each of the three samples shall be recorded and the percent loss in tensile strength computed in the following manner (see 3.7):

$$\% \text{ Loss in Tensile Strength} = \frac{A-B}{A} \times 100$$

where

A = Average Tensile Strength of Control

B = Average Tensile Strength After Immersion

4.5.2.1 Test Results. When test results reflect more loss in tensile strength than specified for the test solution (Table II), it shall be cause for rejection of the lot represented.

4.5.3 Total Weight. Five samples shall be taken each 4 x 6 inches with a tolerance of 1/16 inch. The thickness shall be measured as specified in 4.5.1, and each sample shall be weighed. The weight in grams of each sample shall be computed, and results of the five shall be averaged. The average shall be reported, plus the average (mean) deviation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

*4.5.4 Requirements for Crimped, Cut Filament Polyester. Proportion of the fiber shall be tested in accordance with ASTM D1117-69.

4.6 Certificate of Compliance. The supplier shall submit to the contracting officer a certificate of compliance certifying that the pad shall meet the fiber (see 4.5.4) and adhesive (see 3.3) requirements of this specification.

4.7 Examination of Preparation for Delivery. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine compliance with the requirements of Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Application. The packaging, packing, and marking requirements specified herein apply to direct purchases by or direct shipment to the government.

5.2 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.2.1 Level A or C. Sheets shall be packaged in accordance with the packaging requirements of Specification MIL-P-3816 as specified for sheets.

5.3 Packing. Packing shall be Level A, B, or C as specified (see 6.1).

5.3.1 Level A, B, or C. Sheets packaged as specified in 5.2.1 shall be packed in accordance with the packing requirements of Specification MIL-P-3816 as specified for sheets.

5.4 Marking. In addition to any special marking required by the contract or order, all containers shall be marked in accordance with MIL-STD-129.

6. NOTES

*6.1 Intended Use. The cleaning and polishing pads are intended to be used to clean aircraft with aircraft cleaning solutions and solvents. It is not intended to be used for paint removal operations. It is intended to be attached to a conformable applicator head which attaches to a wooden handle.

*6.2 Ordering Data. Procurement documents should specify the following:

Title, number, and date of this specification

Type

Dimension (see 3.4)

Quantity

Selection of applicable levels of preservation, packaging, packing (see Section 5)

Special marking, if required (see 5.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

*6.3 Revision Notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire contents irrespective of the marginal notations and the relationship to the last previous issue.

SUMMARY: The primary purpose of this revision is to incorporate a pad with a thickness of one inch into the specification. Other changes have been made to clarify some sections of the specification, to update referenced documents, and to correct typographical errors.

Custodian:
Air Force - 84

Preparing activity:
Air Force - 84

Project No. 7920-F235

☆ U. S. GOVERNMENT PRINTING OFFICE: 1975 -603-766/0560

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

WARNER ROBINS ALC (MMSS)
ROBINS AFB, GA 31098

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:54Z
Check the source to verify that this is the current version before use.